

Rafael F Barduzzi

PhD student

rfbarduzzi@gmail.com

Universidade Federal de São Carlos

Botany Department

Washington Luís Highway, Km 235

São Carlos, São Paulo state - Brazil

EDUCATION

- 2024-present **Ph.D., Comparative Biology**, Universidade de São Paulo
Title: Unravelling the tempo and mode of independent radiations in the megadiverse genus *Mimosa* L. (Fabaceae, Mimoseae)
Dissertation advisor: Dr. Leonardo M. Borges
Research supervisor (Abroad): Dr. Daniel S. Moen
- 2018-2023 **B.Sc. in Biology**, Universidade Federal de São Carlos
Undergraduate research: Pollen disparity in the ‘Piptadenia group’ (Leguminosae): differences between lineages and ecological groups
Advisor: Dr. Leonardo M. Borges

SCHOLARSHIPS

- 2025 São Paulo Research Foundation (FAPESP) Research Internship Abroad Scholarship
- 2024-present FAPESP Doctoral degree (Direct) Scholarship
- 2021-2023 FAPESP Undergraduate Research Scholarship

PUBLICATIONS

Barros-Souza, Y., Maianne, M., **Barduzzi, R. F.**, & Borges, L. M. (2026). Beyond the Highlands: Climate Drives Evolutionary Connections Between Ancient Neotropical Mountains and Lowland Biomes. *Global Ecology and Biogeography*, 35(2), e70214.
<https://doi.org/10.1111/geb.70214>

Barduzzi, R. F., Liau-Kang, S., Duarte, A. F. T., Santos, F. D. A. R. D., & Borges, L. M. (2026). Morphological innovation and lineage-specific history drive disparification in the aggregated pollen of mimosoid plants. *Annals of Botany*, 137(1), 159-170
<https://doi.org/10.1093/aob/mcaf213>

Barduzzi, R. F., Barros-Souza, Y., Liau-Kang, S., & Borges, L. M. (2024). Pollen evolution in the Stryphnodendron clade (Leguminosae, Mimoseae). *Brazilian Journal of Botany*, 48(1),

15. <https://doi.org/10.1007/s40415-024-01059-y> (Part of special issue, *Advances in Legume Systematics* 15)

Liau-Kang, S., **Fernandes Barduzzi, R.**, Lozano Calderón, G., Marega Junior, E., Simon, M. F., Ribeiro dos Santos, F. D. A., & Borges, L. M. (2024). More of almost the same: novel pollen descriptions uncover subtle morphological variation in *Mimosa* L. (Leguminosae, Mimosaceae). *Brazilian Journal of Botany*, 47(4), 1071-1090. <https://doi.org/10.1007/s40415-024-01033-8> (Part of special issue, *Advances in Legume Systematics* 15)

MANUSCRIPTS SUBMITTED

Duarte, A. F. T., **Barduzzi, R. F.**, Ribeiro, P. G., Queiroz, L. P., & Santos, F. A. R. *In Press*. Morphological characterization of the pollen grains of species from the new genus *Marlimorimia* and the new circumscription of *Pityrocarpa*.

Barduzzi, R. F., Santos, M., & Borges, L. M. *Under review*. Tree thinking and alternative knowledge among undergraduate Biological Sciences students.

PRESENTATIONS

Talks

- 2026 **Barduzzi, R. F.** A radiation space for *Mimosa*: Evolutionary radiations underlie diversity of one of Brazil's most species-rich plant genera. 2nd Brazilian Congress of Evolutionary Biology, Belo Horizonte (MG), Brazil
- 2026 **Barduzzi, R. F.** Toward a radiation space of the sensitive plants. V Symposium on Comparative Biology, Ribeirão Preto (SP), Brazil
- 2025 **Barduzzi, R. F.** Evolutionary radiations in the genus *Mimosa* L. (*Leguminosae*). UCR Center for Conservation Biology's 7th annual EcoSlam, Riverside (CA), USA
- 2024 **Barduzzi, R. F.** How to identify evolutionary radiations. 11th Meeting on Comparative Biology, Ribeirão Preto (SP), Brazil
- 2023 **Barduzzi, R. F.**; Liau-Kang, S.; Santos, F. A. R. and Borges, L. M. Pollen disparity in the *Mimosa* and *Stryphnodendron* clades (Caesalpinioideae, Mimosoid clade): comparing lineages and ecological groups. 8th International Legume Conference, Pirenópolis (GO), Brazil
- 2022 **Barduzzi, R. F.** and Borges, L. M. Pollen disparity in *Piptadenia* group (Leguminosae), 2022, São Carlos. 28th Universidade Federal de São Carlos Congress on Undergrad Research, São Carlos (SP), Brazil

Posters

- 2025 **Barduzzi, R. F.**; Moen, D. S.; Barros-Souza, Y.; Maianne, M.; Borges, L. The role of leaf morphology, elevation, soil and climate in *Mimosa* L. (Leguminosae) diversification. Botany 2025, Palm Springs (CA), USA
- 2024 **Barduzzi, R. F.**; Barros-Souza, Y.; Maianne, M.; Borges, L. M. Radiations of the sensitive plants: exploring the influence of abiotic niche in diversification of the mega-genus *Mimosa* L. (Leguminosae). 1st Brazilian Congress of Evolutionary Biology, Curitiba (PR), Brazil
- 2023 **Barduzzi, R. F.**; Liao-Kang, S.; Barros-Souza, Y.; Borges, L. M. Evolution of pollen morphology in the Stryphnodendron clade (Fabaceae, Caesalpinioideae). 73th Brazilian Congress of Botany, Belém (PA), Brazil
- 2022 **Barduzzi, R. F.**; Liao-Kang, S. and Borges, L. M. Pollen disparity in the Piptadenia group (Leguminosae): differences between lineages and ecological groups. 2022. 72th Brazilian Congress of Botany, Brazil

TEACHING EXPERIENCE

Academia

- 2025 Phytogeographic Analyses in R, **Short course** at the IV Rio–São Paulo Botany Meeting (*Jornada Rio–São Paulo de Botânica*), co-organizer.
- 2024 Pollen grains: Understanding their morphology and importance for taxonomy, **Workshop** at the XVI Universidade Federal de São Carlos Biology Week (*Semana Acadêmica da Biologia*), Lead organizer
- 2023 Behavioral Biology, **Undergraduate course**, Universidade Federal de São Carlos, Teaching Assistant

Student supervision

- 2026-present Diogo Mantovanelli Rabetti (Universidade Federal de São Carlos – **Undergraduate Research**). Co-supervision with Leonardo M. Borges
- 2025-2026 Felipe Ribeiro Alves Selotto (Universidade Federal de São Carlos – **Bachelor's Thesis**). Co-supervision with Leonardo M. Borges

FIELDWORK

Central Brazil (2023), Southwest Brazil (2024, 2026)

LANGUAGES

Portuguese (native); English (fluent)

PROFESSIONAL SOCIETIES

Brazilian Society of Evolutionary Biology (2024–)
Society for the Study of Evolution (2024–)